



CPAD 504: ECONOMICS AND DATA SCIENCE FOR PUBLIC ADMINISTRATORS

Gateways 2026 Conference, Washington, DC, September 23-26, 2026



SGX3 is funded by the National Science Foundation under award number 2231406.

Oyebade Kunle Oyerinde — Clark Atlanta University — CPAD 504 — FacultyHack@Gateways 2026 September 23-26, 2026

ECONOMICS FOR PUBLIC ADMINISTRATORS

Integrating Economics, Data Analytics, and AI with Jetstream2 and CloudBank for Evidence-Based Policy

A Graduate Course Designed for the Data-Driven Public Sector

COURSE LEVEL
Graduate
3 Credit Hours

OFFERED
Fall & Spring
6:00 pm – 8:30 pm

INSTRUCTOR
Dr. Oyebade Kunle Oyerinde
Full Professor
Special Assistant to the Provost for AI, Data, and Analytics

AFFILIATIONS
2026 NSF NAIRR
AI Education Fellow
Clark Atlanta University

COURSE DESCRIPTION

This course equips public administrators with the essential tools of economics and data science to support evidence-based decision-making and effective policy implementation. Students will integrate economic theory with practical data analysis using R, Python, and Excel, leverage AI-assisted tools, and utilize advanced cyberinfrastructure to evaluate policy options, analyze market outcomes, and communicate insights through compelling data storytelling and visualization.

By the end of the course, students will be prepared to apply economic reasoning and data science methodologies to real-world challenges in the public sector while upholding ethical standards in data and AI use.

KEY TOPICS

- Microeconomics & Macroeconomics for Public Policy
- Data Analysis & Visualization in R, Python, and Excel
- AI-Assisted Policy Analysis & Decision Support
- Market Failures, Public Goods & Policy Solutions
- Ethics, Equity & Responsible Use of Data and AI

ADVANCED COMPUTING RESOURCES

Jetstream2

- NSF-funded national computing resources for research and education
- High-performance cloud environment for data analysis, modeling, and AI
- Supports reproducible and scalable research workflows

CloudBank

- CloudBank Hub provides easy access to computing resources
- Used in the AUC Data Science Initiative pre-freshman program
- Enables accessible, cloud-based learning for all students

SCIENCE GATEWAYS POWERING LEARNING

Seamlessly access Jetstream2, CloudBank, and community resources through science gateways to accelerate discovery, collaboration, and innovation in education and research.

INTEGRATED AI TOOLS FOR LEARNING & RESEARCH

SGX3 Gateways LLMs

Access LLMs via Jetstream gateways for coding, explanations, and insights.

Claude

Generate ideas, summaries, and research sources.

Codex

Generate and refine Python code for data analysis.

AI tools enhance productivity, critical thinking, and innovation in public sector problem-solving.

STUDENT LEARNING OUTCOMES

- Explain and apply economic principles to analyze public policy and organizational decisions.
- Analyze economic data using R, Python, and Excel in cloud environments.
- Communicate insights through data visualization, storytelling, and reports.
- Evaluate market failures and design evidence-based policy solutions.
- Assess ethical considerations in data science and AI applications.

DATASETS FOR ANALYSIS

DOE Open Data Repository
<https://doi.ccs.ornl.gov/datasets>

Access trusted, open datasets for research, teaching, and policy analysis.



ABSTRACT

Science Gateways as Pathways to Inclusive AI Education

Attending Gateways 2026 would strengthen my work as a 2026 NSF NAIRR AI Education Fellow and help expand equitable access to AI and cyberinfrastructure at Clark Atlanta University. As a Full Professor and Special Assistant to the Provost for AI, Data, and Analytics, I am committed to integrating advanced computing resources into teaching, research, and workforce development.

Through my NAIRR Jetstream2 allocation and the use of CloudBank Hub during the Atlanta University Center Data Science Initiative's July 2026 Pre-Freshman Program, I have shown how Science Gateways can provide accessible, cloud-based learning environments for students in AI and data science.

I will also share lessons learned from integrating cloud computing and responsible AI into graduate and undergraduate curricula at an HBCU. I look forward to learning from the Science Gateways community, exchanging ideas on curriculum innovation and faculty development, and broadening participation in AI education. Ultimately, I hope to build collaborations that advance the use of Science Gateways to prepare a diverse next generation of AI researchers and educators.

Jetstream2
CloudBank

NSF NAIRR
National Artificial Intelligence Research Resource

Clark Atlanta University • Gateways 2026 Conference

TARGET COURSE DESCRIPTION	
Course Name	Economics and Data Science for Public Administrators
Course Number	CPAD 504
Department	Public Administration/Sociology and Criminal Justice
Enrollment	20
Prerequisites	None
Project Repo	https://github.com/ooyerind/FacultyHack-Gateways26

COURSE IMPACT

This course advances CAU's mission by leveraging world-class cyberinfrastructure and AI to prepare a diverse next generation of public sector leaders who are data-literate, ethically grounded, and ready to deliver impactful solutions.

NAIRR
National Artificial Intelligence Research Resource

WHERE DISCOVERIES BEGIN

Course Goals

- Identify** and explain the principles and concepts of microeconomics, macroeconomics, and data science that are applicable in decision-making by organizations.
- Analyze** economic data using software programs and programming languages, such as R, Python, and Excel in Jetstream2 and CloudBank Hubs.
- Construct** alternative policy options to remedy a market failure through problem
- Communicate** through storytelling about data visualizations in R and/or Python.

ASSIGNMENTS AND GRADING BREAKDOWN

Problem Sets 1 & 2
= 37.5%

Two problem sets designed to build analytical skills and apply economic and data science concepts using R and data tools.

Exams
= 40%

Comprehensive assessments of economic theory, data analysis, and policy application.

Weekly Discussions
= 15%

Active participation in weekly discussions to exchange ideas, engage with peers, and apply concepts.

Jetstream2/CloudBank Use
= 7.5%

Hands-on use of Jetstream2 and CloudBank resources for computing, analysis, and research activities.

INTRODUCTION

This course prepares public administrators to apply economic principles and data science techniques to make evidence-based decisions and develop effective public policies. By integrating economics with applied data analysis, students gain the analytical skills needed to evaluate policy alternatives, optimize resource allocation, and address complex public sector challenges.

Course Description

This course, *taught in Fall and Spring as needed*, equips public administrators with the essential tools of economics and data science to support evidence-based decision-making and effective policy implementation by integrating macroeconomic and microeconomic theories with applied data analysis. Students will learn to identify and explain economic and data science principles relevant to organizational choices.

Key Content by Week

Module 3: Data Science: R Basics and Data Visualization.

FINAL PROJECT / DELIVERABLE

Problem Sets

- [Step 1 — Start with demand and supply Equations to compute the equilibrium points]
- [Step 2 — Use LLMs to get the R code to graph the results]
- [Step 3 — Produce graphical visualizations of the results]
- [Step 4 — Submit their outputs on CANVAS]
- [Step 5 — Use the grading rubric to assess the outputs]

Jetstream2
CloudBank

CANVAS

FINAL-PROJECT WORKFLOW

- START WITH DEMAND AND SUPPLY EQUATIONS**
Use demand and supply equations to compute the equilibrium points.
- USE LLMs TO GET THE R CODE**
Use Large Language Models (LLMs) to generate or refine R code to create the required graphs.
- PRODUCE GRAPHICAL VISUALIZATIONS**
Run the R code and produce your graphical visualizations of the demand, supply, and equilibrium results.
- SUBMIT THEIR OUTPUTS ON CANVAS**
Use the grading rubric to evaluate student outputs against the following criteria.
- USE THE GRADING RUBRIC TO ASSESS THE OUTPUTS**
Use the grading rubric to evaluate student outputs against the following criteria.

CANVAS

Mentor Feedback Incorporated

- Use of datasets from <https://doi.ccs.ornl.gov/datasets>
- Use of SGX3 Gateways LLMs (<https://llm.jetstream-cloud.org/auth?redirect=%2F>)
- Use of Jetstream2/CloudBank

Economics and Data Science

Made with AI

IMPACT ON STUDENTS

Empowering the next generation of data-literate, AI-ready public sector leaders.

- PRIMARY USERS:** Graduate students in Economics for Public Administrators, with potential extension to undergraduate AI and data science initiatives.
- ADVANCED COMPUTING ACCESS:** Students gain hands-on access to Jetstream2 and CloudBank Hub for cloud-based computing and data analysis.
- TECHNICAL SKILLS:** Students develop practical skills in R, data analysis, data visualization, and AI-assisted research.
- AI LITERACY:** Students learn to use SGX3 Gateways LLMs, Claude, and Codex to support research, generate ideas, identify sources, and develop/present R code.
- AUTHENTIC DATA EXPERIENCE:** Students work with real-world datasets through the DOE Open Data Repository rather than relying solely on hypothetical examples.
- POLICY APPLICATION:** Students apply economic theory, data analysis, and AI tools to evidence-based public policy and decision-making.
- RESPONSIBLE AI:** Students develop awareness of ethical, responsible, and critical use of AI and data.

PREPARING DATA-LITERATE, ETHICALLY-GROUNDED, AI-READY LEADERS TO SOLVE PUBLIC SECTOR CHALLENGES AND IMPROVE COMMUNITIES.

ECONOMICS AND DATA SCIENCE FOR PUBLIC ADMINISTRATORS

Resources and Workflow for Final Project

KEY RESOURCES

Jetstream2
CloudBank

Use Jetstream2 and CloudBank for data analysis in R.

SGX3 Gateways LLMs
<https://llm.jetstream-cloud.org>

Claude
<https://claude.ai>

Codex
<https://chatgpt.com/codex>

DOE.CCS.ORNL
<https://doi.ccs.ornl.gov/datasets>

Science Gateways
<https://www.sciencegateways.org/resources/browse>

FINAL PROJECT WORKFLOW

- START WITH ECONOMIC MODELS**
Use demand and supply equations to compute equilibrium price (P*) and quantity (Q*).
- USE AI TOOLS**
Leverage AI tools (SGX3 Gateways LLMs, Claude, Codex) to generate ideas, find sources, and write or refine R code.
Outputs: R code, Explanations, References
- ANALYZE DATA IN R**
Use Jetstream2 and CloudBank to run R analysis on datasets from DOE.CCS.ORNL.
Outputs: Cleaned data, Analysis results, Tables
- VISUALIZE & INTERPRET RESULTS**
Create visualizations in R and interpret findings.
Outputs: Graphs/Charts, Interpretation, Policy insights
- SUBMIT & EVALUATE**
Submit all deliverables on CANVAS. Instructor assesses using the grading rubric.
Deliverables: R script (R), Output (Tables/graphs), Written interpretation

GRADING RUBRIC AREAS

- Economic Analysis (Equilibrium & Concepts)
- R Code Quality (Correctness & Efficiency)
- Data Visualization (Clarity & Accuracy)
- Interpretation & Policy Insights (Depth & Relevance)
- Ethics & Use of Data (Responsible Use)

PROJECT GOALS

- Reinforce economic concepts:** Apply micro/macro theories to practical public-sector problems.
- Develop analytical skills:** Use data, equations, and quantitative methods to analyze economic policy choices.
- Build R proficiency:** Support the learning on CANVAS. Use R for data analysis, computation, and graphical visualization.
- Integrate AI responsibly:** Use LLMs to support coding, research, and problem-solving while critically evaluating outputs.
- Apply real-world data:** Work with authentic datasets to connect economic theory with evidence-based policy and decision-making.

REFERENCES

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COURSE REDESIGN: BEFORE & AFTER

INITIAL SYLLABUS

- Manual completion of Problem Sets and Exams
- Weakly Discussions
- No use of SGX3 Gateways LLMs
- No use of Jetstream2/CloudBank

AUTHORS & MENTORS

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MODIFIED SYLLABUS

- Use of datasets from <https://doi.ccs.ornl.gov/datasets>
- Use of SGX3 Gateways LLMs (<https://llm.jetstream-cloud.org/auth?redirect=%2F>)
- Use of CloudBank with points toward the final course grade
- Use of R in Jetstream2/CloudBank for data visualization

PROJECT GITHUB REPO

<https://github.com/ooyerind/FacultyHack-Gateways26>

Sponsors: SGX3 • National Science Foundation • ORNL • Omnibond • TACC • HackHPC

SGX3
Extend. Expand. Exemplify.

OAK RIDGE
National Laboratory

Omnibond
Engineering • Trust • Identity

TACC
TEXAS ADVANCED COMPUTING CENTER

HackHPC
Hackathons and Codeathons

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hackhpc.github.io/facultyhack-gateways26